

Ali Ghorbani Ranjbary, DVM-Ph.D.

April 19, 2026

CONTACT INFORMATION

Department of Immunology at University
of Tehran, Iran

Phone: +989 (17) 702 0981

E-Mail
Google Scholar
ORCID

APPOINTMENT

- **Molecular Biology Researcher**, Department of Immunology at University of Tehran, Iran.
- **DVM**, Tabiat company, Tehran. Iran.

RESEARCH BACKGROUND

- **Immunology and tumor microenvironment**
Cell signaling, Data Analysis/Mining, Analysis of tumor microenvironment, Trans-signaling, Apoptosis, Metastasis, Cancer research, Immunosurveillance, DNA mismatch repair.
- **Cloning and engineering**
CRISPR cas13a and 9, T cell engineering, PD1/PDL1 signaling, Transduction (Lentiviral vector), RNA Analysis, Genome research.

EDUCATION

- **Ph.D.**, Ferdowsi University of Mashhad, Mashhad, Iran. GPA: 17.59 out of 20(GPA 4). Ph.D. thesis: Study of the expression profile of cytokines and enzymes involved in the formation of Glycoconjugates in leukocytes and epithelial cells of colorectal cancer, (2015-2020).
- **DVM.**, Islamic azad university of Kazerun, School of Veterinary Medicine, Kazerun. University, Kazerun, Iran. GPA: 16.85 out of 20 (GPA 4). DVM thesis: Assess the prevalence of Clarithromycin resistance point mutations A2142G, A2143G, A2144G H. pylori strains isolated from sheep milk compared with samples isolated from patients referred to hospitals in southern Iran by PCR-RFLP.

HIGHLIGHTS

- Postdoc: Tehran University of Tehran, Tehran, Iran. Design and synthesis of lenti-viruses equipped with Crispr-cas13a system to control PD1 expression and increase P70S6K gene expression in PBMC cells
- Talent Development Centre (TDC), Ferdowsi University of Mashhad, Mashhad, Iran. One of the top young researchers in Iran (2016-2020).
- One of the top young researchers Islamic azad university in Iran (2010-2016).
- The best researcher of Food and safety health in Iran (2012).
- Review activity for Journal of Scientific Reports (2023).
- Review activity for Journal of Advanced Research (2024).

JOURNALS

- J1. Ganjali, M., Rahimi, M., Ramezan, D. et al, **Ghorbani Ranjbary A(Corresponding)**, Effects of enhanced Melissa officinalis essential oil via carrageenan on mitigating aflatoxin b1-induced cytotoxicity in HT-29 and HEK-293 cells. BMC Complement Med Ther 26, 69 (2026). <https://doi.org/10.1186/s12906-025-05228-8>.
- J2. Jamali V, Saleh GK, Bagherzadeh A, Fakhrian M, Rezaeishoorijeh R, Kardani M, Naseri F, **Ghorbani Ranjbary A.** Apoptosis induction via increasing the level of caspases 3/9 and oxidative stress in colon cancer cells exposed to diamond nanoparticles. Toxicology and Environmental Health Sciences. 2026 Feb 24:1-2. <https://doi.org/10.1007/s13530-026-00304-z>.
- J3. **Ghorbani Ranjbary A**, Mehrzad J, Dehghani H, Hosseinkhani S. Impact of IL-17a on apoptosis and mucinosis-related molecules in the microenvironment of colorectal cancer. Archives of Medical Research.

2025 Jul 1;56(5):103220.

- J4. Sadighi, S.A., Abbasian, V., Sabbaghi, S.S. et al, **Ghorbani Ranjbary A**, Identification of novel diagnostic biomarkers in stages I-IV colorectal cancer. Egypt J Med Hum Genet 26, 75 (2025). <https://doi.org/10.1186/s43042-025-00716-5>.
- J5. **Ghorbani Ranjbary A**, Mehrzad J, Rahbar N, Dehghani H. Impacts of some clinicopathodemography and colorectal tissues key cell cycle and mucin stabilizing molecules on the metastasis trend in colorectal cancer patients. Mol Biol Rep. 2023;50(10):8589-8601.
- J6. **Ghorbani Ranjbary A (Corresponding)**, Bagherzadeh, A., Sabbaghi, S.S. et al. Chlorogenic acid induces apoptosis and cell-cycle arrest in colorectal cancer cells. Mol Biol Rep. 2023 Dec;50(12):9845-9857.
- J7. J, Salimi M, Eskandari F, Sahraei M, **Ghorbani Ranjbary A**, Ghasemi R, Hedayati M, Khodagholi F, Eliassi A, Zardooz H. Endoplasmic reticulum stress inhibition ameliorated WFS1 expression alterations and reduced pancreatic islets' insulin secretion induced by high-fat diet in rats. Sci Rep. 2023;13(1):1860.
- J8. **Ghorbani Ranjbary A(Corresponding)**, Saleh GK, Azimi M, Karimian F, Mehrzad J, Zohdi J. Superparamagnetic Iron Oxide Nanoparticles Induce Apoptosis in HT-29 Cells by Stimulating Oxidative Stress and Damaging DNA. Biol Trace Elem Res. 2023;201(3):1163-1173.
- J9. **Ghorbani Ranjbary A**, Mohammad KajiYazdi. A Analysis of TNF- α , IL6, IL1- α , IL-10 and TGF- β 1 Genes Expression in Acute Myeloid Leukemia Patients in Iranian Patients. Basic & Clinical Cancer Research. 2022;14(1):52-8.
- J10. **Ghorbani Ranjbary A**, Mehrzad J, Dehghani H, Abdollahi A, Hosseinkhani S. The study of expression of KI67 and P53 in co-culturing of colon epithelial cell line HT-29 with IL-17A. Journal of Sabzevar University of Medical Sciences. 2022;29(1):141-53.
- J11. **Ghorbani Ranjbary A**, Mehrzad J, Talebkhan Garoussi M, Zohdi J. Long Term Oral Administration of Oregano Essence Effectively Relieves Polycystic Ovarian Rats through Endocrine and Inflammatory Balance. Evid Based Complement Alternat Med. 2022 ;2022:5303583.
- J12. Azimi M, Mehrzad J, Ahmadi E, Orafei M, Aghaie F, Ahmadi A, Rahimi M, **Ghorbani Ranjbary A(Corresponding)**. The Effect of Thymus vulgaris on Hepatic Enzymes Activity and Apoptosis-Related Gene Expression in Streptozotocin-Induced Diabetic Rats. Evid Based Complement Alternat Med. 2022; 2022:2948966.
- J13. Azimi M, Mehrzad J, Ahmadi A, Ahmadi E, Ghorbani Ranjbary A(Corresponding). Apoptosis Induced by Ziziphora tenuior Essential Oil in Human Colorectal Cancer Cells. Biomed Res Int. 2021; 2021:5522964.
- J14. **Ghorbani Ranjbary A**, Mehrzad J, Dehghani H, Abdollahi A, Hosseinkhani S. Variation in blood and colorectal epithelia's key trace elements along with expression of mismatch repair proteins from localized and metastatic colorectal cancer patients. Biological Trace Element Research. 2020;194(1):66-75.
- J15. Ramezani Nowrozani F, **Ghorbani Ranjbary A**. Effects of the hydroalcoholic extract of Artemisia khorassanica on wound healing potential in mice. Comparative Clinical Pathology. 2018;27(4):1079-83.
- J16. Ghorat F, Azizkhani M, Naji SH, **Ghorbani Ranjbary A (Corresponding)**, Histopathological Evaluation of Burdock (Arctium lappa) Root Hydroalcoholic Extract on Wound Healing. Iran Red Crescent Med J. 2017; 19(1):1-6.
- J17. Kooti W, Kafash-Farkhad N, **Ghorbani Ranjbary A (Corresponding)**, Sharafi-Ahvazi N. The effect of celery (Apium graveolens L.) on reproductive parameter in male wistar rat. Avicenna J Phytomed, 2016; 6 (3): 260-265.

- J18. Ramezani Noroozani F, Ojinejad D, **Ghorbani Ranjbary A (Corresponding)**, Effects of Sodium Nitrite on Liver Enzymes and Histological Structure of Liver in Streptozotocin-induced Diabetic Rats. J Mazandaran Univ Med Sci 2016; 26 (144): 171-179.
- J19. Asmarian Sh, Ghorat F, Jaber N, **Ghorbani Ranjbary A (Corresponding)**, Effect of hydroalcoholic leaf extract of Ziziphora tenuior L. on pain in male rats. Journal of Basic and clinical Pathophysiology 2016, 4(2); 17-22.
- J20. **Ghorbani Ranjbary A**, Asmarian Sh, Marhamatizadeh M. Identification of the Prevalence of Resistance to Clarithromycin in Helicobacter Pylori Isolated from Gastric Biopsy via PCR Method. J Babol Univ Med Sci. 2015;17(5):37-43.
- J21. **Ghorbani Ranjbary A (Corresponding)**, Ghorbani Ranjbary N, Varzandian S, Ghorbani-Ranjbar Z. Frequency of Antitoxocara Antibodies in Students with Chronic Cough and Eosinophilia in a Rural Area of Shiraz, Iran, Zahedan Journal of Research in Medical Sciences 2015; 17(1): 15-18.
- J22. **Ghorbani Ranjbary A**, Bahadori R, Moridi R. The Study of Milk Containing Lactobacillus Acidophilus on Histological and Serum Markers of Liver Tissue Injury in Streptozotocin-3. J Babol Univ Med Sci. 2015;17(4):7-13.
- J23. **Ghorbani Ranjbary A**, Varzandian S, Zarei A, Asmarian Sh, Jouibar F. Investigation of Hydroalcoholic Extract of Silybum Marianum on Open Wound Healing in Mice, J Babol Univ Med Sci; 16(5); 2014; pp: 35-41.
- J24. **Ghorbani Ranjbary A**, Ghorbani Ranjbary N, Ghorbani Ranjbary Z, Jouibar F. Effects of Intraperitoneal Injection of Extracts of Origanum Vulgare on Gonadotropin and Testosterone Hormones in Male Wistar Rats, J Babol Univ Med Sci 2014;16(4): 57-63.

Journals (Under Review)

- J1. **Ghorbani Ranjbary A (Corresponding)**. Colorectal Cancer immunity and immunotherapy: Challenges and Future Prospects in Cell and Gene Therapy. **journal of Hematology & Oncology (Review)**. Manuscript ID: c1dc6f68-b1f6-49a8-9732-b48505027e86 v1.
- J2. **Ghorbani Ranjbary A (Corresponding)** Cytokines TNF- α , IL-12p35, IFN-gamma gene expression in the patients with Helicobacter pylori-resistance to antibiotics **Chin Med J (Engl)**. CMJ-D-25-00578R1.

BOOK Chapters

- B1. Zoonosis and ways to prevent it publication a Persian Spring 2011, Number of pages: 248.
- B2. Cat or dog epilepsy, publishing a Community Summer 2011, Number of pages: 69.
- B3. Pre-surgery anesthesia in the horse book, published by a Persian Spring 2013
- B4. Risks associated with general anesthesia in horses, in press, publication Islamic Azad University, Tehran, 2013. Book selection as tuition by the Research Council
- B5. Antibiotics and antibiotic resistance in southern Iran, publication, publication Shahid Beheshti Hospital, Shiraz, 2014.
- B6. Pain and its effective treatment, in press, publication Shahid Beheshti Hospital, Shiraz, 2014.
- B7. Stem cell biology publication a Andishegan, Iran. 2020, Number of pages: 211.
- B8. **General Pathology of Clinical Disease. English, 2022. link: <https://apexpub.info/wp-content/uploads/2023/11/shnaji-apexpub-20231128.pdf>**

CONFERENCES	I have presented 35 papers in various congresses in the form of lectures and posters. For examples: - C1. Ghorbani Ranjbary A, et al. Experimental Effect of plant extracts of mountain on the Thymus enzymes ALT, AST and ALP in adult male Wistar rats, Seventeenth Iranian Veterinary Congress, April 2012: 85-88. - C2. Ghorbani Ranjbary A, et al. Prevalence of genes FnbA, FnbB producing enterotoxin genes in Staphylococcus aureus isolated from cows with subclinical and clinical mastitis city Shiraz, Shiraz Animal Health Management Society 2014 Conference., Pages: 141-145. - C3. Ghorbani Ranjbary A, et al. Study of nettle hydro alcoholic extraction (Urtica dioica) influence on spermatogenesis and Testosterone, spermatogenesis hormones, changes in rats., National Conference molecular cellular News, April 2012; p: 174. - C4. Ghorbani ranjbary A, et al., Antibiotic resistance patterns of Staphylococcus aureus isolated from visitors of patients in hospitals in Shiraz, 2012. 15th International Iranin Congress of Microbiology, School of Medicine, Tehran University of Medical Sciences 26-28 August 2014; p:1043. - C5. Ghorbani ranjbary A, et al., Antimicrobial Resistance and survey of pathogenic genes of stx1 and stx2 in E. coli O157: H7 isolates from beef cattle at industrial slaughter house of fars Province with Multiplex PCR method. 15th International Iranin Congress of Microbiology, School of Medicine, Tehran University of Medical Sciences 26-28 August 2014; p:415. - C6. Ghorbani ranjbary A, et al., Serological assessment of Helicobacter pylori in patients referred to endoscopy department of Fars province hospitals. 15th International Iranin Congress of Microbiology, School of Medicine, Tehran University of Medical Sciences 26-28 August 2014; pp: 593-594.
GRANTS & FUNDINGS	- G1. Design and synthesis of lenti-viruses equipped with Crispr-cas13a system to control PD1 expression and increase P70S6K gene expression in PBMC cells. (IR-TU-1399.12). - G2. Study of the expression profile of cytokines and enzymes involved in the formation of Glycoconjugates in leukocytes and epithelial cells of colorectal cancer. (IR-FU-1395.07.15). - G3. Study on the effects of consumption of milk containing Artemia urmiana and Lactobacillus acidophilus on serum lipid profiles in rats that fed whit high-fat food. (IR-Young Researchers Club-1393.01.25).
Patent	Process for Selective Removal of Heavy Metals (Lead, Mercury, Cadmium, and Arsenic) from Tuna Flesh (Species: Yellowfin and Skipjack) via Ascorbic Acid Complexation Aimed at Producing Canned Tuna with Safety Levels Exceeding International Standards.140550140003003606.
PROFESSIONAL EXPERIENCE	Postdoctoral Research Scholar Tehran University of Tehran, Tehran, Iran. (2020 –2021) Postdoc research: Design and synthesis of lenti-viruses equipped with Crispr-cas13a system to control PD1 expression and increase P70S6K gene expression in PBMC cells
MENTORING & ADVISING	Research Mentoring, Student Supervision & Committee Work 1. Javad Ali Soofiyan, University of Zabol, Zabol, Iran. Induction of apoptosis of HT-29 treated with ethanolic Neem (Azadirachta indica) leaf extract. 2. Masoud Ghazagh, University of Zabol, Zabol, Iran(MSc). Investigating phytochemicals of the essential oil from Mentha longifolia L. and its effects on the expression of apoptosis-related genes in cancer cells (MSc).
SERVICES	• Reviewer for Journal of Advanced Research, Scientific Reports – Nature, Apoptosis, Naunyn-Schmiedeberg's Archives of Pharmacology, Middle East Journal of Cancer, DARU Journal of Pharmaceutical Sciences, Egyptian Journal of Medical Human Genetics.
SKILLS	

Lab Skills

- RT-qPCR
- Western Blotting
- Electrophoresis
- DAPI
- MTT
- Flow cytometry
- SEM and TEM
- IHC
- Bioluminescence
- Tissue Culture – Primary and Cell Line
- Primer design
- Cloning
- Lentiviral and AAV transduction
- Crispr cas9-13 and gRNA design

Programming and Analysis Skills

- Prism – GraphPad 9 (t-test, ANOVA, ROC, Pearson correlation and heat map)
- genex6
- Snap gene
- Graphical abstract
- CLC
- Oligo 7
- Sedeen Viewer
- Protein Purification
- Beacon Designer 8
- BioEdit 7.2
- NGS analysis (RNA-Seq)
- GEO DataSets and Microarray analysis

- **Language**

- Persian: Native
- English: Fluent

- **Professional Affiliations**

- Iran's National Elites Foundation (INEF)
- Office of Talent Guidance, Ferdowsi university